

# Jiaxing Li

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## Education

### Yale University

Postgraduate Associative in McDougal's Lab

New Haven, CT, US

03/2026 – 09/2027

### Yale University

Postgraduate Associative in Laboratory of Physiology of Behavior

New Haven, CT, US

10/2025 – 03/2026

### Harvard University

Visiting student in Harvard Neuromotor Control Lab

Cambridge, MA, US

07/2024 – 11/2024

### Shanghai Jiao Tong University

BS in Biomedical Engineering

Shanghai, China

09/2021 – 06/2025

## Conference

### 35th Annual Computational Neuroscience Meeting, CNS 2026

Halifax, Nova Scotia, Canada

Poster Presentation

07/2026

Event-based machine-learned reduction of biophysically detailed neuron models

### Society For Neuroscience, SFN 2026

Washington, DC

Poster Presentation

11/2026

Machine learning reduction of biophysical models for multi-scale stimulation analysis

## Publication

- J. Li, R.A. McDougal, "Event-based machine-learned reduction of a biophysically detailed neuron model"  
Manuscript in preparation
- J. Li, M.A. Smith, "The diagnostic accuracy of the MMSE and MoCA is preserved across 8 years of repeated administration"  
Manuscript in preparation
- Y. Li<sup>†</sup>, C. Mo<sup>†</sup>, J. Li<sup>†</sup>, P. Yang, J. Wang, H. Yu, S. Hu, P. Zhang, X. Liu, "Temperature Rise Mechanism of Pacemaker Induced by a Time-varying Gradient Magnetic Field" *IEEE Transactions on Instrumentation and Measurement*, DOI: 10.1109/TIM.2024.3485457<sup>†</sup> *equally-contribute*
- Y. Li<sup>†</sup>, J. Li<sup>†</sup>, C. Mo<sup>†</sup>, P. Yang, J. Wang, H. Yu, S. Hu, P. Zhang, X. Liu, "A Quasi-static Methodology for a Fast and Precise Measurement of Medical Implant Magnetic Torque Induced by Static Magnetic Field in MRI" *IEEE Transactions on Instrumentation and Measurement* *under review*
- Liu. X<sup>†</sup>, C. Mo<sup>†</sup>, J. Li<sup>†</sup>, P. Yang, J. Wang, H. Yu, S. Hu, P. Zhang, "Lumped Parameter Model of Human Whole Body Circulatory Loop" *IEEE Access*, DOI: 10.1109/ACCESS.2024.3491112<sup>†</sup> *equally-contribute*

## Acknowledged Research Contributions

- *The Prevalence of Misreporting and Misinterpreting Correlation Coefficients in Biomedical Literature* — Contributed to data collection, investigation, visualization, and manuscript drafting for a systematic review of statistical reporting in biomedical literature, acknowledged for data collection in the manuscript.

# Personal Experience

## McDougal Lab, Yale University

Postgraduate Associate to Prof. Robert McDougal

New Haven, CT

10/2025 – 03/2026

### Machine learning reduction of biophysical model for multiscale stimulation analysis

- Developed an event-driven ML surrogate for biophysical neuron models in NEURON to predict next-spike timing.
- Built recurrent encoders that map recent excitatory and inhibitory events into compact latent representations of neuronal dynamics.
- Achieved F1/AUC of 0.98 and 0.07-ms next-spike-time MAE while preserving long-horizon spiking dynamics.
- Integrated the surrogate with NEURON for scalable simulation across heterogeneous neuron models.

## Dietrich Laboratory of Physiology of Behavior at Yale University

Postgraduate Associate to Prof. Marcelo Dietrich

New Haven, CT

10/2025 – 03/2026

### A deep learning pipeline for 3D brain-wide mapping of local neuronal ensembles in LSFM dataset

- Developed a deep-learning pipeline for 3D brain-wide neuronal activity mapping from LSFM data.
- Analyzed developmental neural activity patterns to study circuit mechanisms of social behavior.

## Neuromotor Control Lab at Harvard University

Visiting Student to Prof. Maurice Smith

Cambridge, MA

07/2024 – 11/2024

### Performance of MMSE and MOCA remains accuracy stability under repeated administrations

- Analyzed longitudinal NACC data to assess MMSE/MoCA diagnostic accuracy across 8 annual administrations.
- Evaluated CN–MCI and CN–dementia classification using ROC-AUC, Youden index, sensitivity/specificity, and bootstrapping analysis.
- Demonstrated stable diagnostic performance over 8 years, with AUC decline significantly below 1% per year.

## Neuroengineering Laboratory at Shanghai Jiao Tong University

Research Assistant to Prof. Shanbao Tong

Shanghai, China

05/2024 – 06/2025

### Dual-brain synchronization mechanism of jointed attention in natural behavior

- Utilized EEG-Hyperscanning technology with controlled variables to simulate real-life parent-child interaction.
- Designed network thresholds and used Phase-Locking Values to construct within-frequency dual-brain networks.
- Demonstrated prefrontal cortex's key role in joint attention engagement with enhanced network connectivity.

## Neural Information Processing Laboratory at Shanghai Jiao Tong University

Research Assistant to Prof. Puming Zhang

Shanghai, China

05/2023 – 05/2024

### Temperature rise mechanism of medical device induced by gradient magnetic field

- Developed an eddy-current and lumped thermal model to characterize MRI gradient-field-induced heating in active implantable medical devices.
- Evaluated the effects of gel thickness, initial temperature, and magnetic-field waveform on implant temperature rise.
- Identified stronger heating under sinusoidal than triangular gradient waveforms and localized peak heating near the device battery.

### Lumped Parameter Model of Human Whole Body Circulatory Loop

- Developed a closed-loop lumped-parameter model of systemic and pulmonary circulation using electrical analogs of cardiovascular dynamics.
- Implemented and solved ODEs in Simulink to reproduce physiological pressure and flow dynamics.
- Simulated normal, heart-failure, and cardiogenic-shock states and modeled aortic stenosis and mitral regurgitation.

## Shanghai Institute of Medical Device Testing

Research Assistant to Dr. Xun Liu

Shanghai, China

05/2023 – 05/2024

### Unsupervised Segmentation Algorithms for Human Brain Stroke Lesions

- leveraged f-AnoGAN and GMVAE models to segment stroke lesions from MRI images
- Optimized model parameters to achieve superior reconstruction quality with demonstrating an SSIM of 0.816 for normal tissue
- Improved segmentation performance over baseline with enhanced SSIM (20%).

## Honors & Awards

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|----------------------------------------------------------------------------------------------|---------|
| Undergraduate Overseas Training Scholarship, Shanghai Jiao Tong University ( <u>Top 1%</u> ) | 2024.11 |
| Undergraduate Merit Scholarship, Shanghai Jiao Tong University (2023 Annual)                 | 2024.09 |
| <u>Second Place</u> in the Guangdong Province Biomedical Engineering Design Competition      | 2024.06 |
| Undergraduate Merit Scholarship, Shanghai Jiao Tong University (2022 Annual)                 | 2023.09 |
| Undergraduate Merit Scholarship, Shanghai Jiao Tong University (2021 Annual)                 | 2022.09 |

## Skills & Interests

**Software skills:** MATLAB, Simulink, R language, LabVIEW, Multisim

**Wet lab skills:** Perfusion, Immunohistochemistry, Stereotaxic surgery, Confocal microscope, Genotyping, Light-sheet fluorescence microscopy, Two-photon fluorescence microscopy...

**Programming Language:** C++, C, Python, Linux

**Language:** English (IELTS: 7.5), Chinese (native)

**Interests:** NeuroAI, Computational Neuroscience, Neural Dynamic, Machine learning, Neurological Disease, Cognitive Neuroscience